NAE14

November 23, 2022

[ ]:
```
import numpy as np
import pandas as pd
from audioop import minmax
from sklearn.neural_network import MLPClassifier # neural network
from sklearn.model_selection import train_test_split
```

[ ]:
```
Data = pd.read_csv('breast_cancer.csv')
data = np.array(Data) #  array
#
X_train, X_test, Y_train, Y_test = train_test_split(data[:,0:29],data[:
 ↪,30],test_size=0.25,random_state=1)  #       25%
#
'''


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'''
#random_state         :
md = MLPClassifier(solver='lbfgs', hidden_layer_sizes=[30,15], random_state=0)
md.fit(X_train,Y_train) #     neural network

score = md.score(X_test,Y_test)  #
print(score)
```

```
0.958041958041958
```

```
/opt/homebrew/Caskroom/miniconda/base/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:559:
ConvergenceWarning: lbfgs failed to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
    https://scikit-learn.org/stable/modules/preprocessing.html
  self.n_iter_ = _check_optimize_result("lbfgs", opt_res, self.max_iter)
```